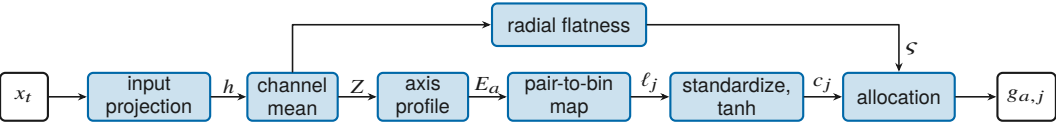
(a) SEGA spectral sensor

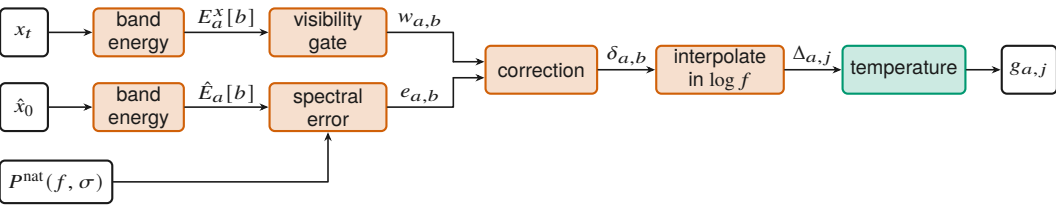


$$Z = \mathrm{mean}_c\, h \qquad K_a = \max(\lfloor L_a/2 \rfloor, 8) \qquad \ell_j = \ln E_a\big((1-\rho_j)(K_a-1)\big)$$

$$\rho_j = j/(D/2-1) \qquad z_j = (\ell_j - \mathrm{mean}_i\, \ell_i)/\mathrm{std}_i\, \ell_i \qquad c_j = \tanh(1.5 z_j) - \mathrm{mean}_i \tanh(1.5 z_i)$$

$$\varsigma = 1 - \mathrm{SF}^{1.5} \qquad m_{\mathrm{ref}} = \max(s,1)^{0.08} \qquad m_{a,j} = \max\big(1,\, m_{\mathrm{ref}}[1 - 0.15\,\varsigma\, c_j]\big),\quad g_{a,j} = m_{a,j}^2$$

(b) Constancy-TM controller



$$\rho_{a,b} = \max\big(E_a^x[b]/(\sigma^2 C) - 1, 0\big) \qquad w_{a,b} = \rho_{a,b}/(1+\rho_{a,b})$$

$$e_{a,b} = \tfrac{1}{2}\big(P_a^{\mathrm{nat}}(f_b,\sigma) + c_a - \ln \hat{E}_a[b]\big) \qquad \delta_{a,b} = \eta\, w_{a,b}\, \mathrm{clip}(e_{a,b}, -1, 1)$$

$$\ln g_{a,j} = \ln \tau + \Delta_{a,j} \qquad \tau = s^{0.16}$$